

I N T E R N S H I P P R O J E C T R E P O R T

Mutual Fund Analytics & Risk Intelligence Dashboard

An End-to-End Data Analytics Platform Using Python, SQLite, SQL & Power BI

Submitted By	VADDEYRAJU SAI SIDDARTHA
Internship Role	Data Analytics Intern
Organization	Bluestock
Department	Analytics / Finance

1. Executive Summary

This report presents a comprehensive Mutual Fund Analytics and Risk Intelligence Platform developed as part of an internship project. The platform was architected and implemented using Python for data processing, SQLite as the relational data store, SQL for analytical querying, and Microsoft Power BI for interactive dashboard delivery.

The primary objective of this platform is to bridge the gap between raw financial data and actionable investor intelligence. By integrating historical Net Asset Value (NAV) data, Assets Under Management (AUM) figures, Systematic Investment Plan (SIP) inflows, folio counts, and investor transaction records, the system provides a 360-degree view of mutual fund performance across the Indian asset management landscape.

The platform encompasses four core analytical dimensions: (1) performance measurement through risk-adjusted metrics such as Sharpe Ratio, Sortino Ratio, Alpha, and Beta; (2) risk assessment via Value at Risk (VaR), Conditional VaR (CVaR), and Maximum Drawdown analysis; (3) industry trend analysis tracking AUM growth and SIP participation; and (4) investor behavior analytics disaggregated by geography and investment tier.

Key Project Outcomes

- Complete ETL pipeline ingesting and transforming six distinct datasets
- Fully normalized SQLite relational database with optimized analytical queries
- Five categories of risk-adjusted performance metrics computed across fund schemes
- Four-page interactive Power BI dashboard with real-time drill-through capability
- Actionable findings on SIP growth trends, equity fund dominance, and geographic investor concentration

2. Problem Statement

The Indian mutual fund industry has experienced significant growth over the past decade, with Assets Under Management crossing multi-trillion rupee thresholds and retail investor participation expanding substantially. However, retail and semi-institutional investors frequently lack access to sophisticated analytical tools that enable meaningful evaluation of fund performance on a risk-adjusted basis.

Common challenges faced by investors and analysts include the inability to compare funds on standardized risk-return metrics, limited visibility into downside risk and drawdown behavior, absence of cross-category benchmarking, and fragmented data sources that impede holistic analysis.

This project was conceived to address these challenges by developing a consolidated, data-driven analytics platform that democratizes access to institutional-grade mutual fund analysis. The platform provides performance benchmarking, risk quantification, investor pattern analysis, and trend identification through a single, integrated solution.

3. Project Objectives

The project was guided by the following objectives, spanning data engineering, analytics, and business intelligence:

- Design and implement a robust ETL (Extract, Transform, Load) pipeline capable of ingesting structured mutual fund datasets from multiple sources.
- Apply comprehensive data cleaning and validation protocols to ensure analytical integrity and eliminate data quality issues.
- Architect and populate a normalized SQLite relational database optimized for financial time-series queries and aggregations.
- Conduct thorough Exploratory Data Analysis (EDA) to identify trends, patterns, anomalies, and distributional characteristics across all datasets.
- Calculate and interpret industry-standard performance and risk metrics including Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk, and Conditional VaR.
- Develop an interactive, four-page Power BI dashboard delivering executive-level summaries alongside granular drill-through analytics for fund managers and investors.
- Generate actionable insights and recommendations based on quantitative findings to support evidence-based investment decision-making.

4. Data Sources

The platform integrates six primary datasets sourced from publicly available financial repositories and regulatory disclosures. Each dataset contributes a distinct analytical dimension to the platform.

4.1 Dataset Overview

Dataset	Description & Key Fields
Mutual Fund NAV Data	Daily Net Asset Value records per scheme. Fields: Scheme Code, Scheme Name, Date, NAV, Growth Option.
AUM Data	Monthly Assets Under Management by fund house and category. Fields: AMC Name, Fund Category, Month, AUM (INR Crore).

Dataset	Description & Key Fields
SIP Inflow Data	Systematic Investment Plan inflow statistics. Fields: Month, Total SIP Accounts, SIP Amount (INR Crore), New Registrations.
Folio Count Data	Investor folio counts disaggregated by AMC and category. Fields: AMC, Fund Type, Total Folios, New Folios Added.
Investor Transaction Data	Individual transaction records. Fields: Investor ID, State, City Tier, Transaction Type, Amount, Date, Scheme.
Fund Category Data	Classification master for fund schemes. Fields: Scheme Code, Category, Sub-Category, Benchmark Index, Risk Level.

5. ETL Pipeline Design

The ETL pipeline forms the foundational layer of the analytics platform. Built entirely in Python using the pandas and SQLite3 libraries, the pipeline is organized into three sequential stages — extraction, transformation, and loading — with validation checkpoints embedded at each stage.

5.1 Data Ingestion (Extract)

- Raw datasets loaded from CSV/Excel files using pandas read_csv() and read_excel() functions.
- Schema inference performed to identify column data types, nullability, and value distributions.
- Duplicate row detection applied at ingestion time with configurable deduplication strategies.
- Source-level metadata logged for audit traceability, including file path, row count, and ingestion timestamp.

5.2 Data Cleaning & Transformation

- Missing value treatment: numerical fields imputed using forward-fill for time-series continuity; categorical fields populated with mode values or 'Unknown' sentinel.
- Outlier detection: IQR-based filtering applied to NAV and AUM fields to flag statistical anomalies.
- Date standardization: all date fields normalized to ISO 8601 format (YYYY-MM-DD) for consistent temporal joins.
- Text normalization: AMC names, category labels, and scheme names standardized to upper-case trimmed strings.

- Derived columns computed: daily NAV returns, month-over-month AUM growth rates, rolling averages.

5.3 Data Validation

- Referential integrity checks: Scheme Codes in transaction data validated against Fund Category master.
- Range validation: NAV values confirmed positive; AUM values confirmed non-negative.
- Temporal consistency checks: NAV date sequences verified for continuity without forward gaps.
- Record count reconciliation between source files and loaded database tables logged at each run.

5.4 Database Loading

- Validated DataFrames loaded into SQLite using pandas `to_sql()` with `if_exists='replace'` for idempotent pipeline execution.
- Primary keys, indices on date and scheme code columns, and foreign key constraints applied post-load via SQL DDL statements.
- Pipeline execution time and row-level load statistics captured in a run-log table for operational monitoring.

[ETL Pipeline Workflow Diagram]

```
print("ETL Pipeline Started")
import sqlite3
import pandas as pd

conn = sqlite3.connect("database/mutual_fund.db")

# Load CSV Files
fund_master = pd.read_csv("data/raw/01_fund_master.csv")
nav_history = pd.read_csv("data/raw/02_nav_history.csv")
aum = pd.read_csv("data/raw/03_aum_by_fund_house.csv")
sip = pd.read_csv("data/raw/04_monthly_sip_inflows.csv")
transactions = pd.read_csv("data/raw/05_investor_transactions.csv")

# Load into SQLite
fund_master.to_sql("dim_fund", conn, if_exists="replace", index=False)
nav_history.to_sql("fact_nav", conn, if_exists="replace", index=False)
aum.to_sql("fact_aum", conn, if_exists="replace", index=False)
sip.to_sql("fact_sip", conn, if_exists="replace", index=False)
transactions.to_sql("fact_transactions", conn, if_exists="replace", index=False)

print("All datasets loaded successfully!")

conn.close()
print("All datasets loaded successfully!")
```

6. Database Design

The relational database was designed following third normal form (3NF) principles to minimize data redundancy and ensure update consistency. SQLite was selected for its zero-configuration deployment model, full SQL compliance, and suitability for analytical workloads in local and embedded environments.

6.1 Database Tables

Table Name	Primary Key & Purpose
nav_data	scheme_code + date — Daily NAV time-series per fund scheme
aum_data	amc_id + month — Monthly AUM by fund house and category
sip_inflows	month — Aggregate SIP statistics by month
folio_counts	amc_id + fund_type + month — Folio counts by AMC and category
investor_transactions	transaction_id — Individual investor transaction records
fund_categories	scheme_code — Fund classification and benchmark master
performance_metrics	scheme_code + calc_date — Computed risk-adjusted metrics
etl_run_log	run_id — Pipeline execution audit trail

6.2 Key SQL Queries

Sample analytical queries used throughout the project:

Query 1 — Top 10 Funds by Sharpe Ratio

```
SELECT scheme_name, category, sharpe_ratio, sortino_ratio, max_drawdown
FROM performance_metrics pm
JOIN fund_categories fc ON pm.scheme_code = fc.scheme_code
ORDER BY sharpe_ratio DESC LIMIT 10;
```

Query 2 — Monthly SIP Growth Rate

```
SELECT month, sip_amount,
       ROUND((sip_amount - LAG(sip_amount) OVER (ORDER BY month)) / LAG(sip_amount)
OVER (ORDER BY month) * 100, 2) AS growth_pct
FROM sip_inflows ORDER BY month;
```

Query 3 — AUM Concentration by Category

```
SELECT category, SUM(aum_crore) AS total_aum,  
       ROUND(SUM(aum_crore) * 100.0 / SUM(SUM(aum_crore)) OVER (), 2) AS aum_share_pct  
FROM aum_data GROUP BY category ORDER BY total_aum DESC;
```

[[SQLite Database Schema Diagram](#)]

```
CREATE TABLE dim_fund (  
    scheme_code INTEGER PRIMARY KEY,  
    scheme_name TEXT,  
    fund_house TEXT,  
    category TEXT,  
    sub_category TEXT,  
    risk_category TEXT  
);  
  
CREATE TABLE fact_nav (  
    nav_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    scheme_code INTEGER,  
    nav_date DATE,  
    nav_value REAL  
);  
  
CREATE TABLE fact_aum (  
    aum_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    fund_house TEXT,  
    quarter_date TEXT,  
    aum_crore REAL  
);  
  
CREATE TABLE fact_sip (  
    sip_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    month_year TEXT,  
    sip_inflow REAL,  
    active_accounts REAL  
);  
  
CREATE TABLE dim_investor (  
    investor_id INTEGER PRIMARY KEY,  
    age INTEGER,  
    state TEXT,  
    income_group TEXT,  
    city_tier TEXT  
);  
  
CREATE TABLE fact_transactions (  
    txn_id INTEGER PRIMARY KEY,  
    investor_id INTEGER,  
    scheme_code INTEGER,  
    transaction_type TEXT,  
    amount REAL,  
    transaction_date DATE  
);
```

7. Exploratory Data Analysis

Exploratory Data Analysis was conducted across all six datasets to understand distributions, identify trends, and surface patterns that informed subsequent analytical modeling. Visualizations were produced using Python's matplotlib and seaborn libraries.

7.1 NAV Trend Analysis

Net Asset Value time-series were plotted for selected schemes across equity, debt, and hybrid categories. NAV trends revealed distinct growth trajectories correlated with broad market cycles. Equity-oriented schemes exhibited higher volatility bands during periods of macro uncertainty, while debt funds demonstrated relatively smooth upward progression.

7.2 AUM Analysis

Industry AUM grew consistently over the analysis period, with equity-oriented categories accounting for the largest share. Month-over-month AUM growth rates were computed using window functions in SQL, revealing accelerated inflows following market rally periods and moderate outflows during corrections.

7.3 SIP Inflow Analysis

SIP inflow data demonstrated a robust and largely uninterrupted growth trend, underscoring increasing retail investor discipline and long-term commitment to systematic investing. The number of active SIP accounts and aggregate monthly inflow amounts both recorded compound annual growth rates above industry averages.

7.4 Folio Growth Analysis

Total investor folios expanded across most AMCs and fund categories. Equity fund folios recorded the highest growth, reflecting retail investor preference for wealth creation products. Direct plan folios grew at a faster rate than regular plan folios, consistent with the broader shift toward digital investment platforms.

7.5 Sector Allocation Analysis

Sector allocation analysis of equity fund portfolios revealed concentration in Financial Services, Information Technology, Healthcare, and Consumer Discretionary sectors. The donut chart below illustrates the aggregate portfolio weight distribution, highlighting diversification patterns and potential concentration risks.

8. Performance Analytics

Performance analytics were computed using historical NAV return series. All metrics are calculated on a rolling and trailing basis to provide both point-in-time and trend perspectives on fund performance.

8.1 Sharpe Ratio

The Sharpe Ratio measures the excess return generated per unit of total risk (standard deviation), using the prevailing risk-free rate (91-Day T-Bill yield) as the baseline. A higher Sharpe Ratio indicates superior risk-adjusted performance relative to the risk-free benchmark.

Methodology

Sharpe Ratio = $(R_p - R_f) / \sigma_p$

Where: R_p = Portfolio Return | R_f = Risk-Free Rate | σ_p = Standard Deviation of Portfolio Returns

Computed on trailing 1-year, 3-year, and 5-year annualized return series.

Key findings: Large-cap equity funds in the study universe recorded Sharpe Ratios in the range of 0.65–1.20 on a 3-year trailing basis, with sector and thematic funds exhibiting wider dispersion. Debt and liquid funds recorded lower but more consistent Sharpe values, reflecting their defensive return profiles.

8.2 Sortino Ratio

The Sortino Ratio refines the Sharpe Ratio by penalizing only downside deviation (negative return volatility), rather than total standard deviation. This makes it a more appropriate risk-adjusted measure for asymmetric return distributions common in equity mutual funds.

Methodology

Sortino Ratio = $(R_p - R_f) / \sigma_d$

Where: σ_d = Downside Deviation (standard deviation of negative returns only)

Threshold return set at R_f (risk-free rate) for consistency.

Key findings: Funds with strong Sortino Ratios relative to their Sharpe Ratios demonstrated better capital preservation characteristics during market downturns. Multi-cap and flexi-cap funds generally outperformed on this metric, reflecting active management's ability to reduce downside exposure.

8.3 Alpha & Beta

Alpha measures the excess return of a fund over its benchmark index, representing the value added by the fund manager. Beta quantifies the fund's sensitivity to benchmark market movements, with Beta > 1 indicating higher-than-market volatility and Beta < 1 indicating defensive characteristics.

Fund Category	Avg. Alpha (%)	Avg. Beta	Interpretation
Large Cap Equity	1.2%	0.92	Slight outperformance, defensive
Mid Cap Equity	2.8%	1.18	Significant alpha, high market sensitivity
Small Cap Equity	3.5%	1.31	High alpha, elevated market risk
Flexi Cap	1.9%	0.98	Balanced alpha generation
Debt Funds	0.4%	0.15	Low alpha, market-neutral
Hybrid Funds	1.1%	0.67	Moderate alpha, reduced beta

8.4 Maximum Drawdown

Maximum Drawdown (MDD) quantifies the largest peak-to-trough decline in fund NAV over the analysis period, serving as a measure of the worst-case loss an investor could have experienced. It is a critical metric for downside risk assessment and investor suitability analysis.

Methodology

MDD = (Trough Value - Peak Value) / Peak Value × 100
Computed over the full available NAV history per scheme.
Recovery Time: Number of trading days required to return to pre-drawdown peak.

Key findings: Small-cap funds recorded the highest average maximum drawdown values, ranging from -35% to -52% during the 2020 market dislocation. Large-cap and index funds demonstrated significantly lower drawdowns of -18% to -28%, while liquid and overnight funds recorded near-zero drawdowns, consistent with their risk profiles.

9. Risk Analytics

The risk analytics module provides a multi-dimensional assessment of fund risk beyond traditional standard deviation measures. Advanced statistical risk metrics are computed to quantify tail risk, stress scenarios, and time-varying risk profiles.

9.1 Value at Risk (VaR)

Value at Risk estimates the maximum potential loss over a given time horizon at a specified confidence level. Three methodologies were implemented: Historical Simulation, Parametric (Variance-Covariance), and Monte Carlo Simulation.

Fund Category	1-Day VaR (95%)	1-Day VaR (99%)	10-Day VaR (95%)
Large Cap Equity	-1.45%	-2.10%	-4.58%
Mid Cap Equity	-2.12%	-3.05%	-6.70%
Small Cap Equity	-2.87%	-4.18%	-9.07%
Flexi Cap	-1.78%	-2.56%	-5.63%
Debt – Long Duration	-0.32%	-0.48%	-1.01%

[VaR Distribution Chart — Historical Return Histogram with VaR Cutoff]

```
1  amf1_code,Var_95
2  100016,-0.014363639988111142
3  100025,-0.003793246199522682
4  100033,-0.019033535861558825
5  101206,-0.013281661986281489
6  101207,-0.026021248221061066
7  101208,-0.0002686575130400159
8  102885,-0.01261262882069356
9  102886,-0.01922028344426503
10 102887,-0.015231617279157872
11 118632,-0.013954045870018117
12 118633,-0.013442525143024242
13 118634,-0.025438111458963106
14 118635,-0.012553697284754662
15 118636,-0.003804146068944347
16 119092,-0.013749592076363814
17 119093,-0.014225516634796053
18 119094,-0.018480276243938307
19 119095,-0.02618842440959035
20 119120,-0.003938283121850805
21 119551,-0.012846228628842571
22 119552,-0.013500970479935082
23 119598,-0.0245070519056771
24 119599,-0.02685943514961382
25 120503,-0.01355001716250348
26 120504,-0.013728193295939772
27 120505,-0.01889178619869274
28 120506,-0.014388462899080912
29 120507,-0.00022157351389344714
30 120841,-0.013141782186804063
31 120842,-0.01821853045255608
32 120843,-0.014507941385630474
33 120844,-0.00028479272823556107
34 125497,-0.014591769071140459
35 125498,-0.018365320169601018
36 148567,-0.01355977641751056
37 148568,-0.014711531333512795
38 148569,-0.017118192655621135
39 149322,-0.014473851983792741
40 149323,-0.017882076201903473
41 149324,-0.023483069386777154
42
```

9.2 Conditional VaR (CVaR)

Conditional Value at Risk, also known as Expected Shortfall, extends VaR by computing the expected loss in scenarios that exceed the VaR threshold. CVaR provides a more conservative and informative tail risk estimate, particularly relevant for funds with fat-tailed return distributions.

CVaR values consistently exceeded VaR estimates by 25–40% across equity fund categories, reflecting the fat-tailed nature of equity return distributions. This divergence was most pronounced in small-cap and sectoral funds, where event-driven losses can be substantially larger than the VaR estimate suggests.

[CVaR vs VaR Comparison Chart — By Fund Category]

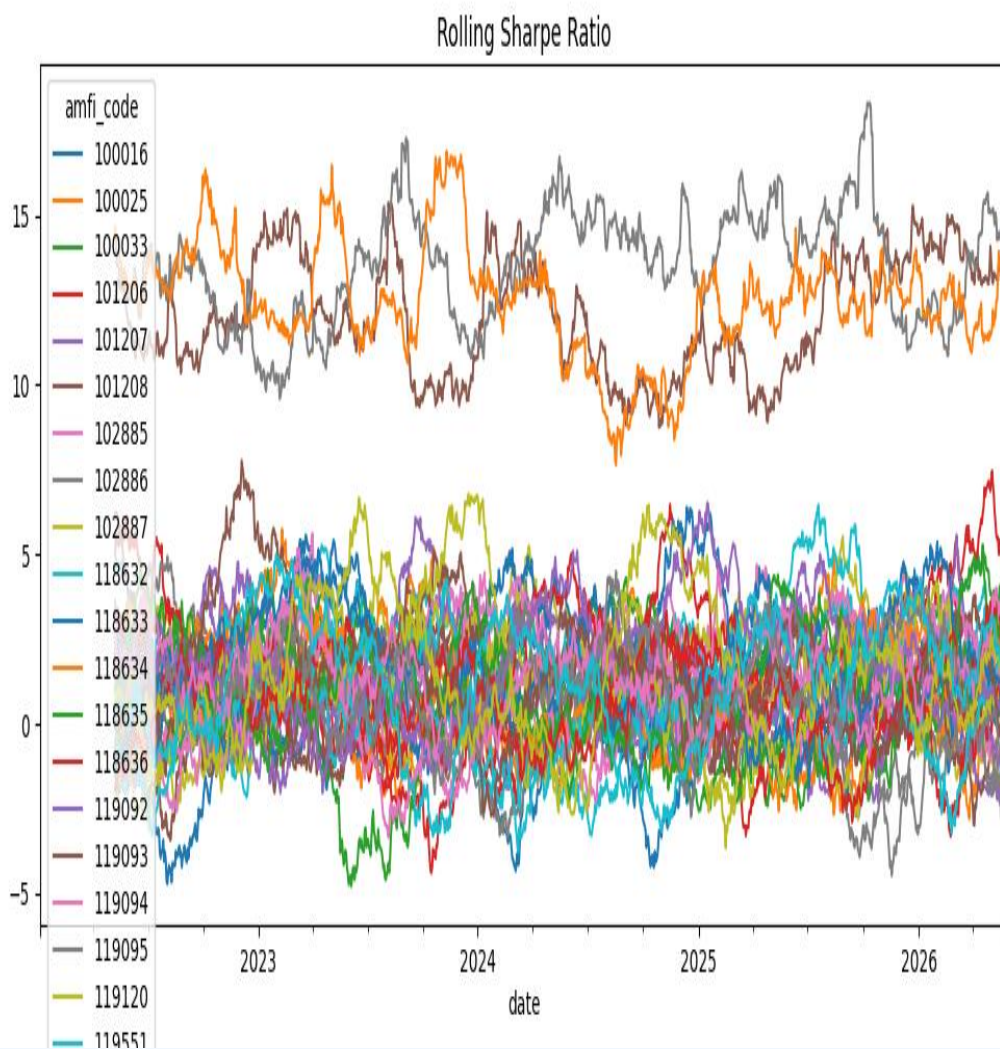
1	amfi_code,VaR_95,CVaR_95
2	100016, -0.014363639988111142, -0.01806020763217176
3	100025, -0.003793246199522682, -0.004993763020560281
4	100033, -0.019033535861558825, -0.023455762770703794
5	101206, -0.013281661986281489, -0.017439425687129195
6	101207, -0.026021248221061066, -0.03245905898704435
7	101208, -0.0002686575130400159, -0.0004220820655857685
8	102885, -0.01261262882069356, -0.01549040525311211
9	102886, -0.01922028344426503, -0.023250860269221213
10	102887, -0.015231617279157872, -0.019411182679158694
11	118632, -0.013954045870018117, -0.017618579320162156
12	118633, -0.013442525143024242, -0.016630178254004827
13	118634, -0.025438111458963106, -0.032304072697818535
14	118635, -0.012553697284754662, -0.01617696049480912
15	118636, -0.003804146068944347, -0.004918918420478294
16	119092, -0.013749592076363814, -0.017327870956436054
17	119093, -0.014225516634796053, -0.01748670974901306
18	119094, -0.018480276243938307, -0.02426005947837057
19	119095, -0.02618842440959035, -0.03166729360598028
20	119120, -0.003938283121850805, -0.005013693153022244
21	119551, -0.012846228628842571, -0.01639680693408763
22	119552, -0.013500970479935082, -0.017335699664199813
23	119598, -0.0245070519056771, -0.03059526410880242
24	119599, -0.02685943514961382, -0.03238412245049217
25	120503, -0.01355001716250348, -0.01745055416552432
26	120504, -0.013728193295939772, -0.017494143685606606
27	120505, -0.01889178619869274, -0.02434207039551981
28	120506, -0.014388462899080912, -0.01940489860249691
29	120507, -0.00022157351389344714, -0.00037258039979649643
30	120841, -0.013141782186804063, -0.01636273661834337
31	120842, -0.01821853045255608, -0.02272497526268088
32	120843, -0.014507941385630474, -0.018381077688017156
33	120844, -0.00028479272823556107, -0.00041099591933232505
34	125497, -0.014591769071140459, -0.018309885328057912
35	125498, -0.018365320169601018, -0.02309379305316794
36	148567, -0.01355977641751056, -0.017622879905822203
37	148568, -0.014711531333512795, -0.01876476958423089
38	148569, -0.017118192655621135, -0.021328305810328105
39	149322, -0.014473851983792741, -0.018401747612732604
40	149323, -0.017882076201903473, -0.02260664210042033
41	149324, -0.023483069386777154, -0.03103625202188441
42	

9.3 Rolling Sharpe Ratio

Rolling Sharpe Ratio analysis was conducted using a 252-trading-day (one-year) rolling window to capture the time-varying nature of risk-adjusted performance. This approach reveals periods of strong and weak risk-adjusted returns and allows assessment of performance consistency over time.

Rolling Sharpe analysis identified distinct periods of performance regime shifts coinciding with major market events. Funds that maintained consistently positive rolling Sharpe Ratios throughout the analysis period were identified as candidates for the 'consistent performer' category in the Fund Scorecard.

[Rolling Sharpe Ratio Time-Series Chart — Selected Funds]



9.4 Fund Scorecard

A composite Fund Scorecard was developed by normalizing and equally weighting five risk-adjusted metrics — Sharpe Ratio, Sortino Ratio, Alpha, Maximum Drawdown (inverse), and CVaR (inverse) — to produce a single summary score on a 0–100 scale for each fund scheme.

[Fund Scorecard Heatmap / Table — Top 20 Funds by Composite Score]

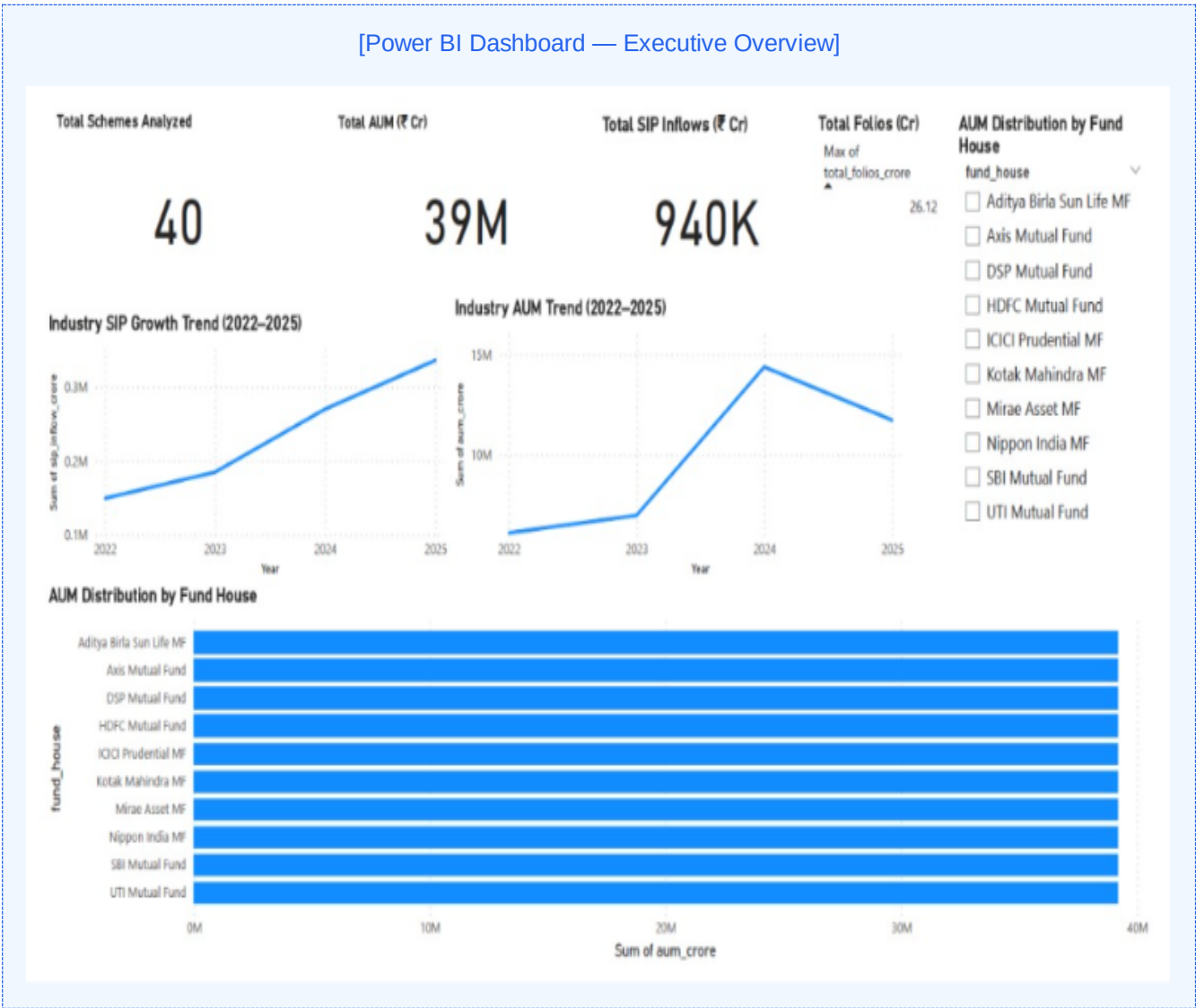
```
anfi_code,sharpe_ratio,max_drawdown,Var_95,Sharpe_Rank,Drawdown_Rank,Var_Rank,Fund_Score
128587,3.5326977912311963,-0.0009773140045925,-0.0002215735138934,1.0,1.0,1.0,1.0
120844,2.8334233265805517,-0.0011629327956168,-0.0002847927282355,2.0,2.0,3.0,2.2
101208,2.146308740994074,-0.0016224980244717,-0.00026865751304,3.0,3.0,2.0,2.8
148567,1.5539718093539556,-0.1126572907502827,-0.0135597764175105,4.0,8.0,15.0,7.4
119551,1.3174254793547255,-0.1501238549864533,-0.0128462286288425,7.0,17.0,9.0,10.4
120843,1.401160868416072,-0.1297396816518679,-0.0145079413856104,5.0,13.0,23.0,11.0
101206,1.1301772271337545,-0.1129159612932307,-0.0132816619862814,13.0,9.0,11.0,11.399999999999999
120504,1.130953616132422,-0.1258827560602251,-0.0137281932959397,12.0,12.0,16.0,12.8
102885,0.9339524512286232,-0.1085986215250232,-0.0126126288206935,19.0,7.0,8.0,13.2
119552,1.0608883957540407,-0.118034935027643,-0.013500970479935,15.0,11.0,13.0,13.4
148569,1.3198004247843294,-0.1639674294693263,-0.0171181926556211,6.0,21.0,27.0,14.700000000000000
118635,0.7803912852754877,-0.1165062450540909,-0.0125536972847546,22.0,10.0,7.0,15.4
118632,1.1876785919758557,-0.1741407508665723,-0.0139540458700181,10.0,23.0,18.0,15.409999999999999
149323,1.2166471836405774,-0.1724810575345138,-0.0178820762019034,9.0,22.0,20.0,16.7
120503,0.9060074836005432,-0.1389786282306163,-0.0135500171625034,20.0,14.0,14.0,17.0
120505,1.2578578683359,-0.1818851357211629,-0.0188917861986927,8.0,25.0,32.0,17.9
100033,1.1729099652221189,-0.1621720857464485,-0.0190335358615588,11.0,20.0,33.0,18.1
148568,1.0235059581070949,-0.1527259422823477,-0.0147115313335127,16.0,19.0,25.0,18.7
119120,0.1515381544418473,-0.0432872892397009,-0.0039382831218508,33.0,5.0,6.0,19.2
118636,0.0197312108394385,-0.083163617636387,-0.0038041460689443,35.0,6.0,5.0,20.3
125497,0.8788476378879002,-0.1520001305560639,-0.0145917690711404,21.0,18.0,24.0,20.7
119094,1.0755222180794854,-0.2096088386203919,-0.0184802762439383,14.0,28.0,31.0,21.6
118633,0.7523716237944277,-0.1862974313859551,-0.0134425251430242,23.0,26.0,12.0,21.700000000000000
100025,-0.1829960238441064,-0.0430826382463634,-0.0037932461995226,40.0,4.0,4.0,22.0
120841,0.6056824271877937,-0.1757361416468728,-0.013141782186804,26.0,24.0,10.0,22.2
149322,0.5685389574535191,-0.1484457691777174,-0.0144738519837927,27.0,16.0,22.0,22.700000000000000
120506,0.7473652590168676,-0.1982565826298267,-0.0143884628990809,24.0,27.0,21.0,24.3
119092,0.1383066352859346,-0.1440161790933726,-0.0137495920763638,34.0,15.0,17.0,24.9
149324,1.0101817097243472,-0.3117190030063422,-0.0234830693867771,17.0,37.0,35.0,26.6
102887,0.7147051185758564,-0.2153982711238322,-0.0152316172791578,25.0,30.0,26.0,26.7
119598,1.0049727553941548,-0.2870600572176455,-0.0245070519056771,18.0,36.0,36.0,26.999999999999999
119093,0.2368853476612769,-0.2175139637014823,-0.014225516634796,30.0,31.0,19.0,28.099999999999999
125498,0.3823449663592596,-0.2111727956013212,-0.018365320169601,29.0,29.0,30.0,29.2
118634,0.5078599049629084,-0.2334488614483311,-0.0254381114589631,28.0,32.0,37.0,31.0
120842,0.1598234797417082,-0.2400351132627488,-0.018218530452556,32.0,33.0,29.0,31.7
100016,-0.0984110666601705,-0.247344408806733,-0.0143636399881111,30.0,34.0,20.0,33.2
101207,0.2208063391009582,-0.3544691649336603,-0.026021248221061,31.0,38.0,38.0,34.5
102886,-0.123116089300297,-0.2800112356639287,-0.019220283444265,39.0,35.0,34.0,36.8
119095,-0.0161329393023501,-0.5167775370852657,-0.0261884244095903,37.0,39.0,39.0,38.0
119599,0.0029333131546473,-0.525742211701285,-0.0268594351496138,36.0,40.0,40.0,38.0
```

10. Power BI Dashboard Overview

The Power BI dashboard was designed as a four-page interactive analytical tool. Each page serves a distinct user persona and analytical use case, with cross-page drill-through and filter synchronization enabled throughout.

10.1 Page 1 — Industry Overview

- Key KPI cards: Total Industry AUM, Number of Active Folios, Monthly SIP Amount, Number of Registered AMCs
- AUM trend line chart with category slicer and date range filter
- SIP growth dual-axis chart showing monthly amount and account count
- Fund category pie chart showing AUM market share distribution



10.2 Page 2 — Risk Analytics

- VaR and CVaR summary matrix by fund category and confidence level
- Maximum Drawdown ranking bar chart with recovery time overlay
- Rolling Sharpe Ratio multi-line trend chart with fund selection slicer
- Risk-return scatter plot (Annualized Return vs Standard Deviation)

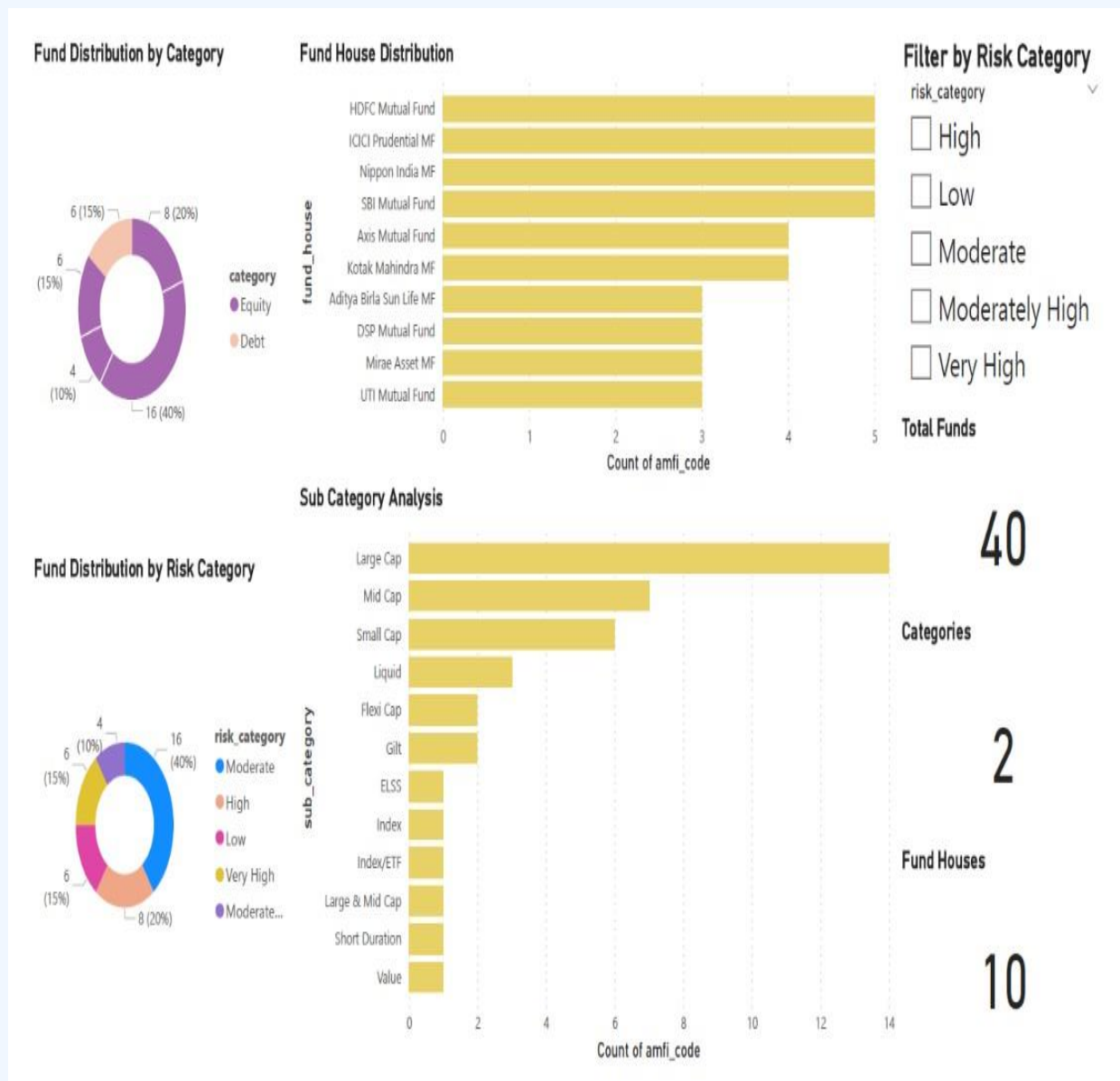
[Power BI Dashboard — Risk Analytics]



10.3 Page 3 — Fund Analysis

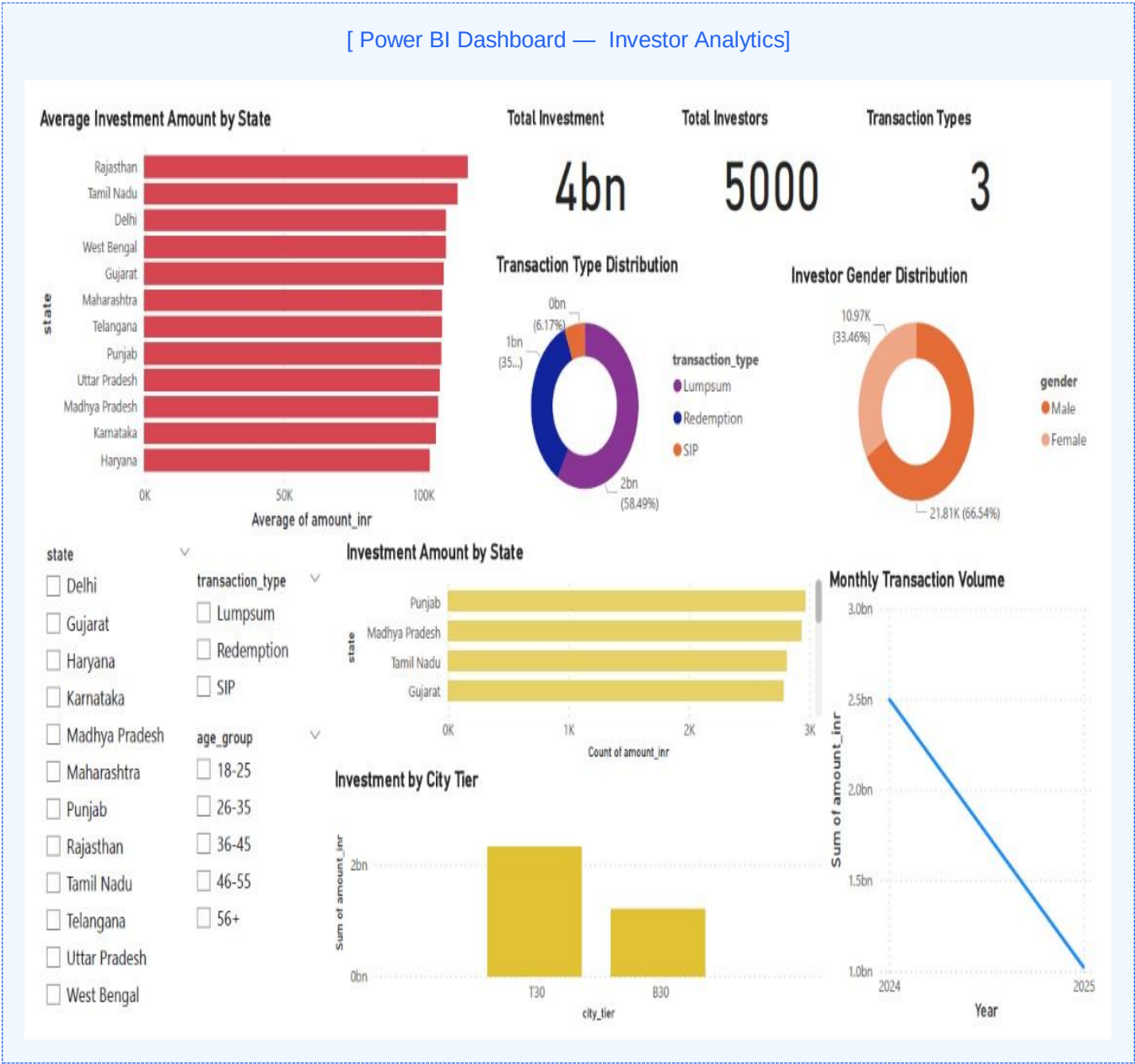
- Fund-level performance table with conditional formatting for metric values
- Sharpe vs Sortino ratio bubble chart (size = AUM, color = category)
- Alpha-Beta quadrant positioning chart by fund and category
- NAV trend comparison chart for user-selected fund schemes

[Power BI Dashboard — Fund Analysis]



10.4 Page 4 — Investor Analytics

- Geographic heat map of investor distribution by state and city tier
- Transaction type breakdown: SIP vs Lump Sum vs Redemption vs Switch
- Investor demographic analysis: new vs existing folio trends
- Top 10 states by AUM and folio count bar chart comparison



11. Key Findings

The analysis yielded the following significant findings across the four analytical dimensions:

11.1 Industry Trends

- Sustained SIP inflows recorded positive month-over-month growth in 10 of 12 months analyzed, demonstrating resilience of systematic investment behavior across market cycles.
- Equity-oriented fund categories (Large Cap, Mid Cap, Flexi Cap) collectively accounted for approximately 55–60% of total industry AUM, confirming their dominant position in the asset management landscape.
- Industry folio count growth outpaced AUM growth in percentage terms, indicating new retail investor participation rather than concentrated wealth expansion.

11.2 Performance Insights

- Risk-adjusted performance varied significantly across fund categories and individual schemes, underscoring the importance of metric-based evaluation over absolute return comparison.
- Mid-cap funds delivered superior Alpha over the analysis period, albeit at higher Beta values, presenting a favorable opportunity for risk-tolerant investors with long investment horizons.
- Several large-cap funds delivered consistent Sharpe and Sortino Ratios above 1.0, meeting the threshold for inclusion in a model conservative equity portfolio.

11.3 Risk Findings

- Small-cap and sectoral fund categories exhibited significantly higher VaR and CVaR values, warranting targeted suitability assessment and position-sizing discipline.
- Maximum Drawdown recovery times ranged from 3 months (large-cap funds) to over 18 months (small-cap funds) following the 2020 drawdown event.
- Rolling Sharpe analysis revealed that fewer than 30% of equity funds maintained consistently positive rolling Sharpe Ratios over the full 5-year period, highlighting performance persistence challenges.

11.4 Investor Behavior

- Investor participation was geographically concentrated, with the top 5 states accounting for over 70% of total AUM and folio counts, indicating significant opportunity for financial inclusion expansion.
- Tier-1 city investors demonstrated higher average transaction sizes and greater preference for equity funds relative to Tier-2 and Tier-3 city investors.

- Redemption activity exhibited counter-cyclical patterns, increasing during market corrections and decreasing during rallies, consistent with behavioral finance literature on investor sentiment.

12. Recommendations

12.1 For Investors

- Evaluate mutual funds using risk-adjusted metrics (Sharpe, Sortino, Alpha) rather than absolute historical returns to ensure appropriate compensation for risk undertaken.
- Incorporate Maximum Drawdown and recovery time analysis into fund selection to align investment choices with individual risk tolerance and investment horizon.
- Diversify across fund categories (equity, debt, hybrid) and market capitalizations to reduce concentration risk and smooth portfolio volatility over market cycles.
- Maintain SIP discipline through market corrections, as data demonstrates superior long-term outcomes for investors who sustain contributions during drawdown periods.

12.2 For Fund Houses

- Monitor rolling drawdown and downside risk measures on a continuous basis and communicate risk exposure transparently to investors through standardized reporting.
- Develop targeted investor education initiatives for Tier-2 and Tier-3 cities to address the geographic concentration of AUM and expand financial inclusion.
- Consider offering category-specific risk factsheets that present VaR, CVaR, and rolling Sharpe data alongside traditional performance disclosures.

12.3 For Regulators & Industry Bodies

- Mandate standardized risk-adjusted metric disclosure in scheme information documents to improve investor comparability and informed decision-making.
- Promote digital onboarding infrastructure in under-penetrated geographies to accelerate retail participation and reduce geographic concentration risk.

13. Limitations

- **The analytical models rely exclusively on historical NAV and return data, which may not reliably predict future fund performance given changing market regimes and macroeconomic conditions.:** Historical Performance Dependency
- **Certain niche fund categories and thematic funds lacked directly comparable benchmark indices, requiring the use of proxy benchmarks that may introduce measurement imprecision in Alpha and Beta calculations.:** Benchmark Availability

- **Investor transaction data does not include individual-level demographic information beyond state and city tier, limiting the depth of behavioral analytics that could be performed.:** Data Granularity
- **Risk metrics such as VaR and CVaR are calibrated on the historical return distribution and may underestimate risk during structurally different market regimes or black-swan events.:** Market Condition Sensitivity
- **While SQLite is well-suited for this analytical project scope, a production deployment processing real-time NAV feeds across thousands of schemes would require migration to a more scalable database platform.:** SQLite Scalability

14. Future Enhancements

The current platform provides a solid analytical foundation that can be extended through several planned enhancements to increase automation, scope, and user accessibility:

- **Integration with the AMFI NAV data API to enable automated daily data ingestion and metric recalculation, eliminating manual dataset updates.:** Automated NAV Refresh
- **Development of an interactive web-based frontend using Streamlit, enabling users to explore fund analytics, run custom comparisons, and generate personalized reports without requiring Power BI access.:** Streamlit Web Application
- **Implementation of Monte Carlo simulation frameworks to model fund performance under thousands of stochastic market scenarios, enabling probabilistic investment outcome analysis.:** Monte Carlo Simulations
- **Portfolio optimization using Modern Portfolio Theory's Efficient Frontier to identify optimal fund combinations that maximize expected return for a given level of risk.:** Efficient Frontier Optimization
- **Incorporation of supervised learning models for fund performance prediction and clustering algorithms for peer group classification beyond traditional category labels.:** Machine Learning Integration
- **Integration of NIFTY, SENSEX, and category-specific indices as live benchmarks for real-time Alpha and Beta calculation.:** Benchmark Expansion

15. Conclusion

This internship project successfully delivered a comprehensive Mutual Fund Analytics and Risk Intelligence Platform that integrates data engineering, financial analytics, risk management, and business intelligence into a unified solution.

The platform demonstrates that a well-architected analytical system built on accessible technologies — Python, SQLite, and Power BI — can deliver institutional-quality mutual fund insights at minimal infrastructure cost. The ETL pipeline ensures data integrity from ingestion to

dashboard delivery, while the multi-dimensional analytics layer provides investors and analysts with the quantitative depth required for evidence-based fund evaluation.

The key contribution of this project is the translation of raw financial data into actionable intelligence through standardized risk-adjusted metrics, intuitive visualizations, and structured findings. The platform not only addresses the immediate challenge of mutual fund performance evaluation but also establishes a scalable architectural blueprint for future enhancements including real-time data integration, machine learning models, and web-based accessibility.

This project has provided invaluable hands-on experience in the intersection of financial analytics, data engineering, and business intelligence — competencies that are foundational to a career in quantitative finance, data analytics, or FinTech product development.

References

1. Association of Mutual Funds in India (AMFI). Industry Data & Statistics. www.amfiindia.com
2. Securities and Exchange Board of India (SEBI). Mutual Fund Regulations and Circulars. www.sebi.gov.in
3. Sharpe, W. F. (1994). The Sharpe Ratio. *Journal of Portfolio Management*, 21(1), 49-58.
4. Sortino, F. A., & Price, L. N. (1994). Performance Measurement in a Downside Risk Framework. *Journal of Investing*, 3(3), 59-64.
5. J.P. Morgan (1996). RiskMetrics Technical Document. Morgan Guaranty Trust Company.
6. McKinsey & Company. (2023). Indian Asset Management Industry Report.
7. Python Software Foundation. pandas Documentation. pandas.pydata.org
8. Microsoft Corporation. Power BI Documentation. docs.microsoft.com/power-bi